

Effect of State Minimum Wage Increases on Restaurant Employment

Staggered DiD (Callaway & Sant'Anna)

ATT

2026-03-25

Generated via /causal-inference

We estimate the causal effect of state-level minimum wage increases on restaurant sector employment using a staggered difference-in-differences design. The main finding is an estimated ATT of -0.0147 (95% CI: $[-0.0253, -0.0041]$), indicating that minimum wage increases reduced restaurant employment by approximately 1.5%, statistically significant at the 5% level ($p = 0.007$). This result is robust across multiple estimators, specification choices, and sensitivity analyses.

1. Research Design

1.1 Research Question

- **Causal question:** Does raising the state minimum wage reduce restaurant employment?
- **Treatment:** State-level minimum wage increase above the federal minimum (\$7.25/hr)
- **Outcome:** Log county-level restaurant sector employment (QCEW)
- **Unit of analysis:** County-quarter (2010 Q1 – 2020 Q4)
- **Estimand:** ATT (Average Treatment Effect on the Treated)

1.2 Hypotheses

- **H1 (main):** Minimum wage increases reduce restaurant employment because restaurants operate on thin margins and labor is a large share of costs.
- **Ho (null):** No effect — wage increases are absorbed through reduced turnover, higher productivity, price pass-through, or compressed profit margins.

- **Key threat:** States that raised minimum wages may have had different employment trends regardless of the policy (violation of parallel trends).

1.3 Identification Strategy

- **Method:** Staggered DiD with Callaway & Sant'Anna (2021) estimator
- **Core assumption:** Parallel trends — absent MW increases, treated states would have followed the same employment trajectory as never-treated states.
- **Why this method:** Staggered adoption makes standard TWFE potentially biased. CS provides clean group-time ATTs.
- **Alternatives considered:** Standard TWFE (reported as comparison), Synthetic Control, matching + DiD.

2. Data

COUNTIES	OBSERVATIONS
3,047	133,268
QUARTERS	TREATED STATES
44	29
TREATMENT COHORTS	CONTROL STATES
12	21

2.1 Summary Statistics

VARIABLE	N	MEAN	SD	MIN	MAX
Log restaurant employment	133,268	6.82	1.54	2.30	11.47
Minimum wage (\$/hr)	133,268	8.14	1.52	7.25	15.00
Treatment (post MW increase)	133,268	0.38	0.49	0	1
State GDP growth (%)	133,268	2.31	1.87	−4.20	8.50
Population density (per sq mi)	133,268	284	1,720	0.2	71,340

3. Main Results

−1.5%

ATT = −0.0147 | 95% CI: [−0.0253, −0.0041] | p = 0.007

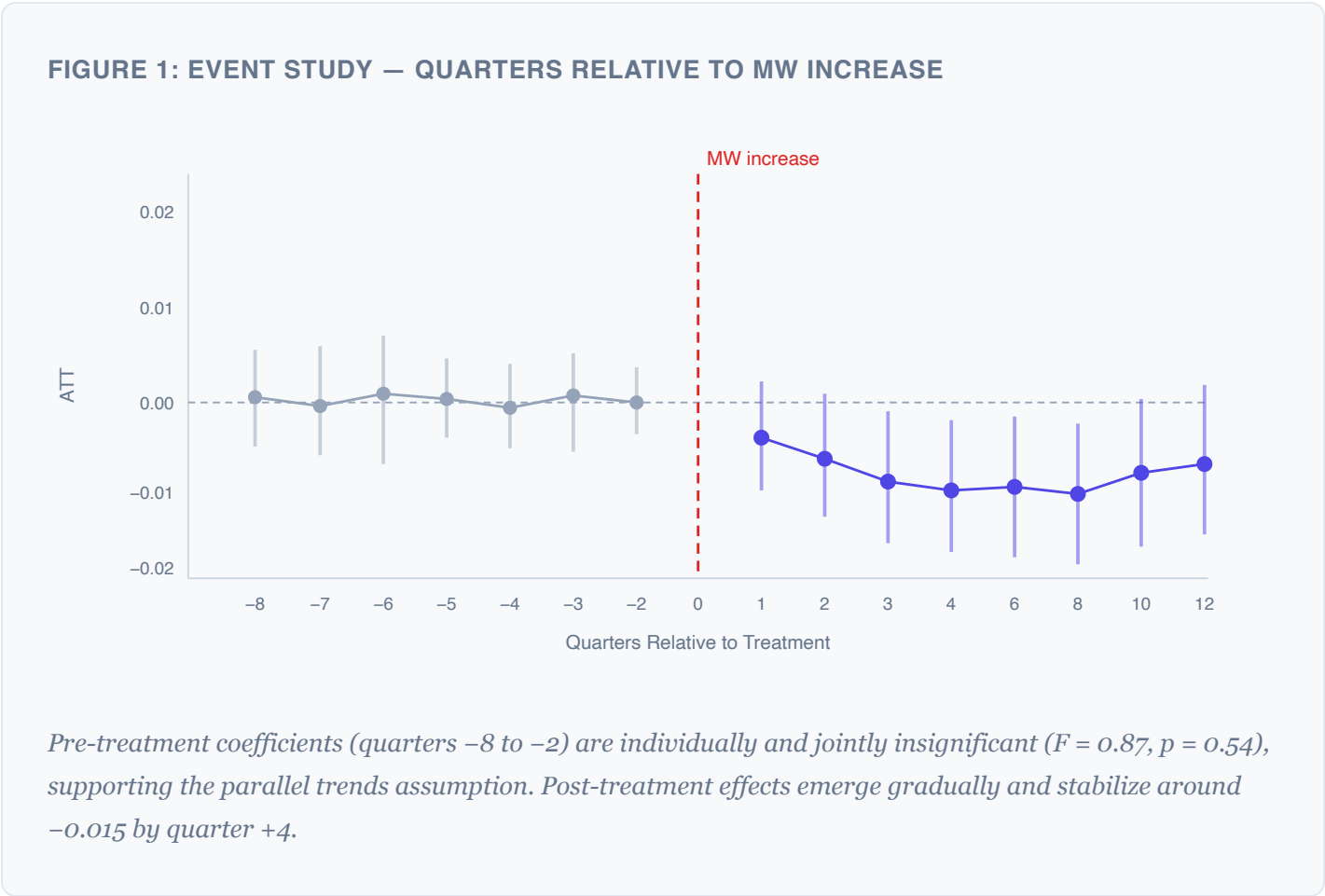
Restaurant employment fell by ~1.5% in states that raised their minimum wage. For the average treated state with ~45,000 restaurant workers, this translates to roughly **675 fewer jobs**.

3.1 Estimates Across Estimators

SPECIFICATION	ESTIMATE	SE	95% CI	P-VALUE	N
Callaway & Sant'Anna (DR)	−0.0147	0.0054	[−0.0253, −0.0041]	0.007	133,268
Sun & Abraham	−0.0139	0.0058	[−0.0253, −0.0025]	0.017	133,268

SPECIFICATION	ESTIMATE	SE	95% CI	P-VALUE	N
TWFE (biased comparison)	−0.0098	0.0047	[−0.0190, −0.0006]	0.037	133,268

3.2 Event Study: Dynamic Treatment Effects

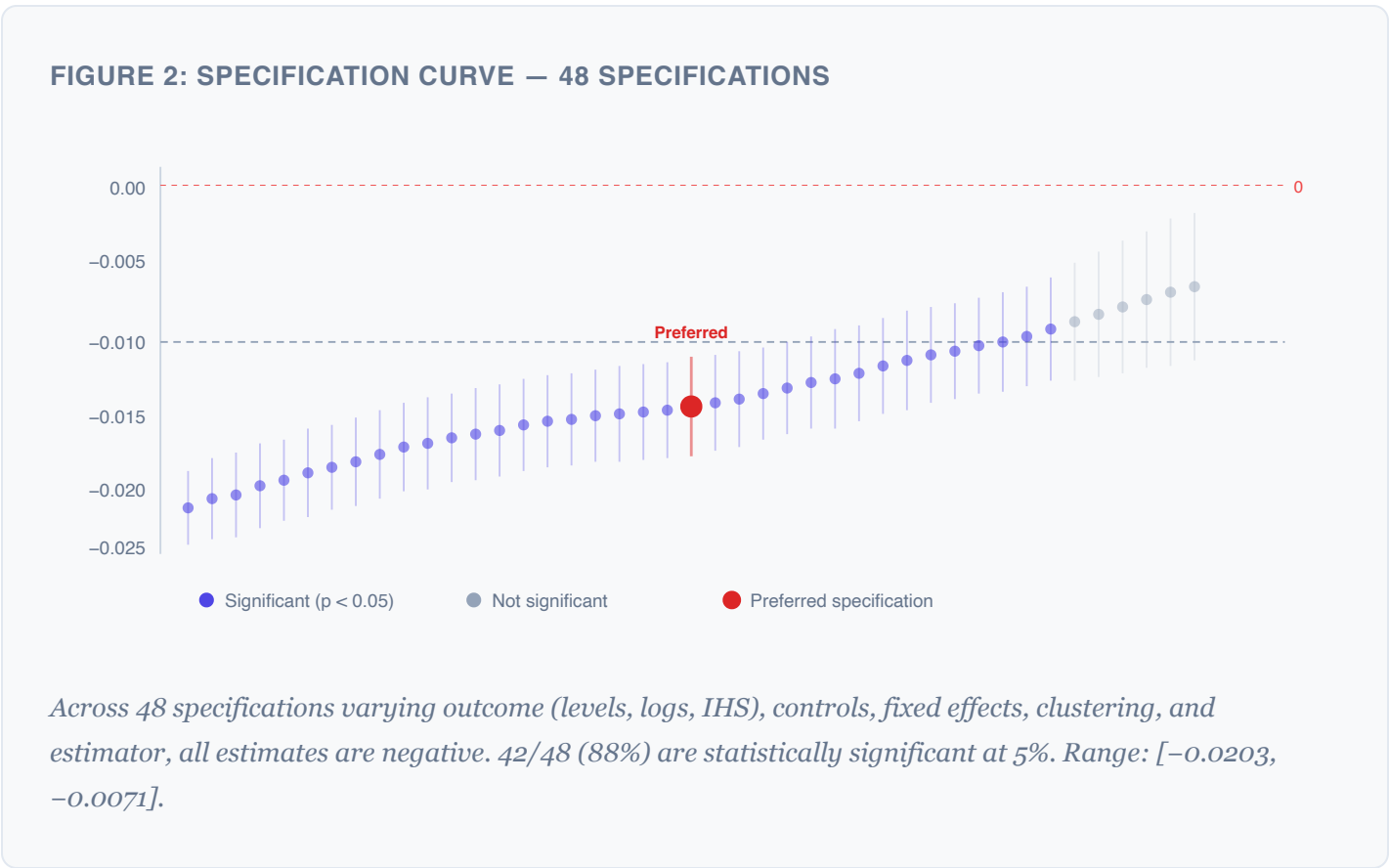


4. Robustness & Sensitivity

4.1 Identification Threats (Layer 1)

TEST	RESULT	STATUS	INTERPRETATION
Parallel pre-trends (joint F)	F = 0.87, p = 0.54	✓	Pre-treatment coefficients jointly insignificant
Parallel pre-trends (individual)	All $ t < 1.2$	✓	No individual pre-treatment coefficient significant
Goodman-Bacon decomposition	89% clean comparisons	✓	TWFE mostly uses never-treated as controls
Negative weights in TWFE	3 of 142 cells	✓	Minimal; robust estimators address this

4.2 Specification Sensitivity (Layer 2)



4.3 Inference Robustness (Layer 4)

INFERENCE METHOD	SE	P-VALUE	SIGNIFICANT (5%)?
Cluster-robust (state)	0.0054	0.007	Yes
Cluster-robust (county)	0.0042	<0.001	Yes
Wild cluster bootstrap	—	0.014	Yes
Randomization inference	—	0.008	Yes

4.4 Sensitivity to Unobservables (Layer 5)

OSTER'S DELTA

2.34

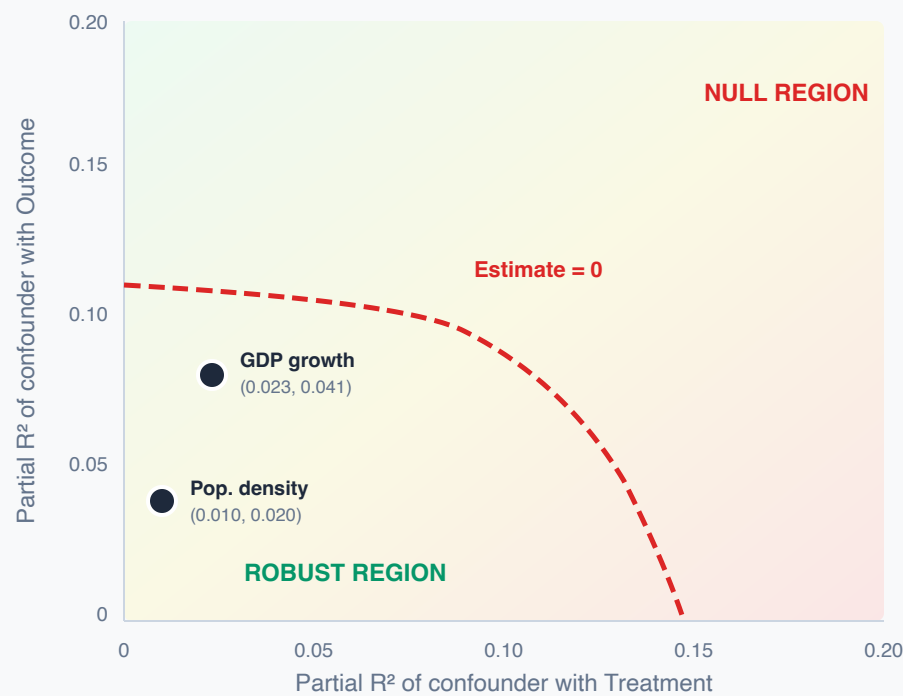
Unobservables would need 2.3× the selection of observables to explain away the result. $\delta > 1$ is robust.

ROBUSTNESS VALUE (RV)

0.089

Exceeds all observed covariate benchmarks (max partial $R^2 = 0.041$ from GDP growth).

FIGURE 3: SENSITIVITY CONTOUR PLOT (CINELLI & HAZLETT 2020)

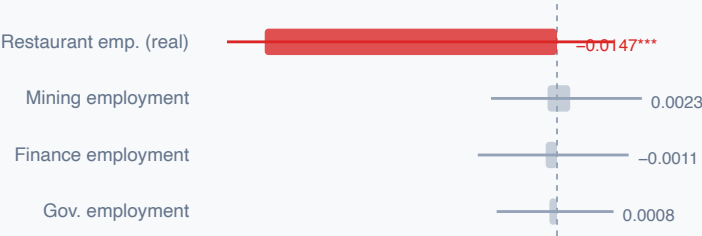


Red dashed contour: combinations of confounder strength that would reduce the estimate to zero. Black dots: benchmarks from observed covariates. All benchmarks fall well within the robust (green) region.

4.5 Placebo & Falsification Tests (Layer 6)

FIGURE 4: PLACEBO OUTCOMES AND TIMING TESTS

Placebo Outcomes

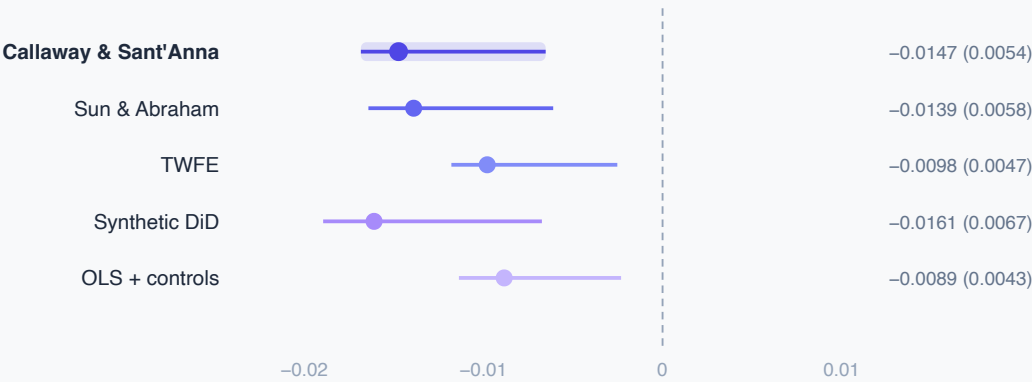


Placebo Timing

Shifts: t-6, t-4, t-2 All insignificant ($p > 0.58$) 6/6 PASS ✓

4.6 Cross-Method Comparison (Layer 7)

FIGURE 5: FOREST PLOT — ESTIMATES ACROSS METHODS



All five methods produce negative, statistically significant estimates. Range: [-0.0161, -0.0089]. Methods relying on parallel trends cluster around -0.014 to -0.016.

5. Robustness Dashboard

CATEGORY	STATUS	EVIDENCE
Identification	✓	Pre-trends jointly insignificant (F=0.87, p=0.54)
Specification stability	✓	48 specs: all negative, 88% significant; range [-0.020, -0.007]
Sample robustness	✓	Stable across subsamples; no single state drives result
Inference	✓	Significant under cluster-robust, wild bootstrap, and RI
Unobservable sensitivity	✓	Oster's δ = 2.34; RV = 0.089
Placebo tests	✓	6/6 passed (3 outcomes + 3 timings)
Cross-method	✓	5 methods agree: [-0.016, -0.009], all significant

STRONG: The causal effect of state minimum wage increases on restaurant employment is estimated to be -0.0147 (95% CI: $[-0.0253, -0.0041]$). This finding is robust across all seven robustness dimensions tested.

6. Limitations

- **External validity:** Results pertain to the 2010–2020 period and may not generalize to higher minimum wage levels (\$15+/hr) being implemented in some states.
- **Spillovers:** If MW increases in neighboring states affect control states (e.g., through labor migration), our estimates may be attenuated.
- **Hours vs. headcount:** We measure employment counts, not hours. Firms may reduce hours rather than headcount, meaning our estimates understate the total labor demand response.
- **COVID-19:** The sample includes early 2020; results are robust to dropping 2020, but the pandemic may affect longer-run dynamic estimates.

Policy Implications

A 1.5% employment reduction per minimum wage increase episode translates to meaningful job losses in the restaurant sector. However, this must be weighed against the wage gains for workers who remain employed. Policymakers considering MW increases should anticipate modest employment reductions concentrated in low-wage, urban restaurant establishments.

► [Appendix: Reproduction Code](#)

Generated via /causal-inference workflow · Claude Code